APPENDIX A –MR SITE VIBRATION TEST GUIDELINES

A–1 TEST MEASUREMENTS

- Vibration measurements are in the range of 10^{-6} g. Test equipment must have the required sensitivity to these levels.
- All analyses are to be narrowband Fast Fourier Transforms (FFT's) over the frequency bands listed in Table A–1.

TABLE A–1
FREQUENCY BANDS FOR FFT'S

FREQUENCY BAND	FREQUENCY RESOLUTION
0.2 to 50 Hz	Δf= 0.125 Hz

- Time histories of the vibration must be recorded (i.e. acceleration levels vs. time). The resolution of the time history must be adjusted to clearly capture the transient events. The analyzer set–up will be site dependent.

A–2 EQUIPMENT (SPECTRAL ANALYZER) SET–UP

- Frequency average a minimum of 20 linear averages (do not use peak hold or 1/3 octave analysis)
- Hanning window must be applied to the entire spectra

Spectrum analyzers capable of these measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer. Accelerometers must have the capability to measure from 0.2 Hz beyond 50 Hz. Time histories can be recorded using any of the analyzers mentioned above. Good quality strip chart recorders may also be sufficient. Please note that the equipment mentioned are for example only. It is the responsibility of the Engineering test firm to provide equipment that will follow the test requirements.

A–3 DATA COLLECTION

A–3–1 Ambient Baseline Condition:

All of the measurements defined in Section A–1 and A–2 must be made in a ‘quiet’ environment. That is, in areas where excessive traffic, subway trains, etc. exists, a vibration measurement must also be made during periods without traffic or during periods of light traffic. Measurements must define the lowest levels of vibration possible at the site.

The source of any steady state vibration whose levels exceed the specifications in ***Direction 2294003 Signa TwinSpeed Pre–Installation*** **Section 4–15–3 Specifications For 1.5T LCC (Active Shield) Magnets** must be identified as to the source of the vibration disturbance. A second measurement should be made with all of the identified contributors powered down if possible. In situations where it is not possible to power down equipment, vibration data must be collected to identify specific source of the vibration concern. The majority of steady state vibration problems can be negated by isolating the vibration source.

A–3–2 Normal (Operating) Condition

All of the measurements listed above must be repeated during periods of ’normal’ environmental conditions including the FFT’s and time histories. The transient measurements must be provided to define the dynamic disturbances the MR system might be exposed to. This transient disturbance is required for a true assessment of the site.

Transient vibration is difficult to assess and to eliminate. Environments exceeding the 500 micro–g, zero to peak trigger level, requires analysis of the building’s dynamic response to the transient event. The building response must remain below the vibration specifications in ***Direction 2294003 Signa TwinSpeed Pre–Installation*** **Section 4–15–3 Specifications For 1.5T LCC (Active Shield) Magnets**. The dynamic response following the transient event must exceed the specifications for sustained periods of time, i.e. for random dynamic response of duration less than a few seconds will have less impact on system performance than response of longer duration.

Building structures excited by transient events exceeding the specifications in ***Direction 2294003 Signa TwinSpeed Pre–Installation*** **Section 4–15–3 Specifications For 1.5T LCC (Active Shield) Magnets** either provide recommendations to reduce levels to meet specifications or an alternative location should be identified. A second vibration measurement should be made to help identify a more stable location.

A–4 PRESENTATION/INTERPRETATION OF RESULTS

The recommended format for site vibration data collection, presentation, and analysis is illustrated in the examples shown in Illustrations A–1 through A–2. Presentation of the data in any other format (linear units only) may result in an incorrect interpretation and diagnosis of the site. Additional data collection or presentation methods is at the option of the vibration testing service.

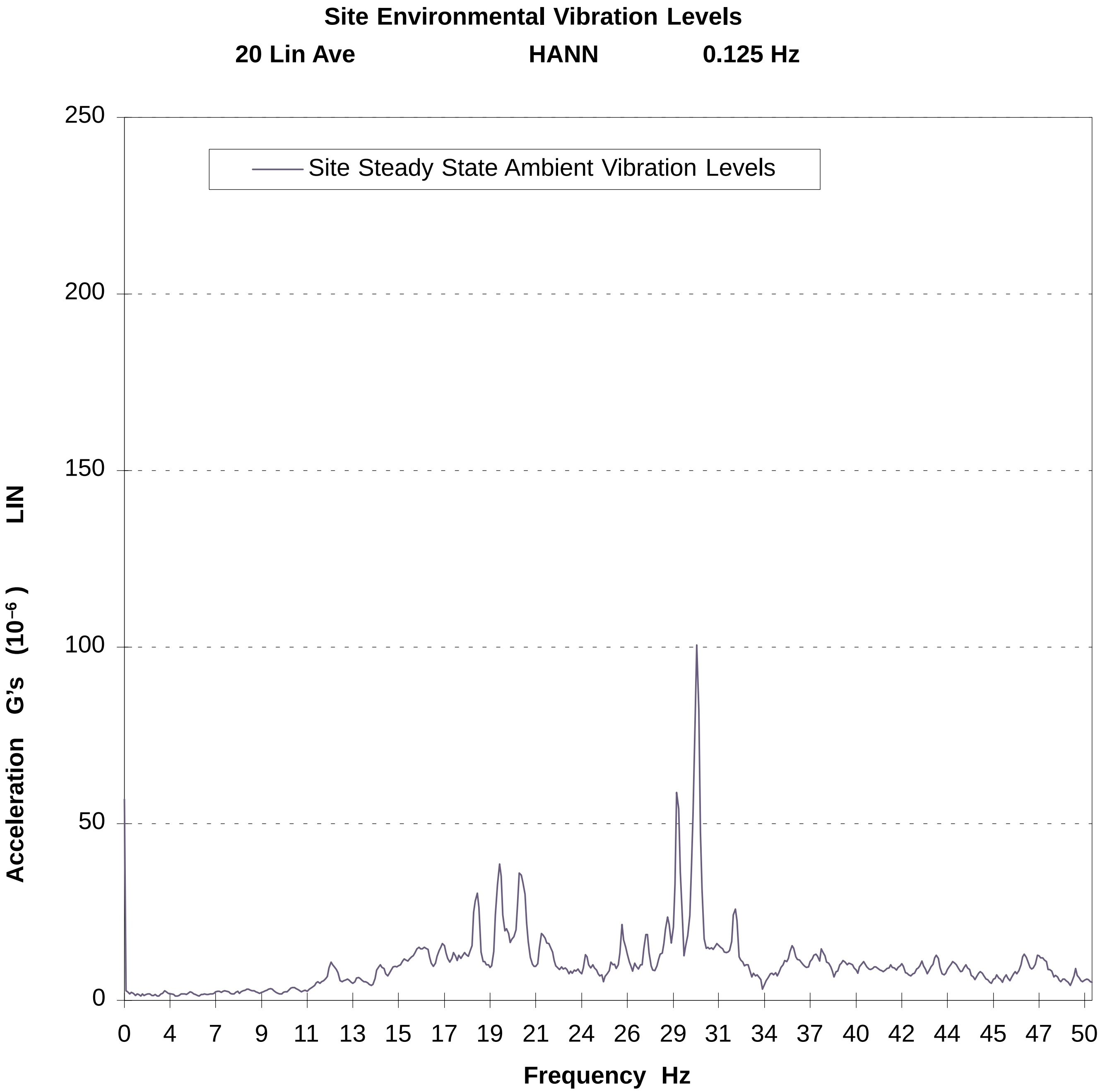
It is the responsibility of the customer's vibration testing service to interpret the results and determine if that site meets GE's specifications. Illustrations A–1 and A–2 are examples provided to assist a test consultant in the use of GE Steady State specifications (vibration specifications above ambient baseline). If the vibration levels are too high, additional data acquisition may be necessary to:

- determine the source of the vibration
- propose a solution to the problem
- find an alternate site location.

Illustrations A–3 and A–4 are examples provided to assist a test consultant in the use of GE Transient specifications. The 500 micro–g, zero to peak trigger level identifies data collection to begin assessment of the site vibration analysis. The response of the transient must be assessed relative to the Steady State vibration specifications in ***Direction 2294003 Signa TwinSpeed Pre–Installation*** **Section 4–15–3 Specifications For 1.5T LCC (Active Shield)**.

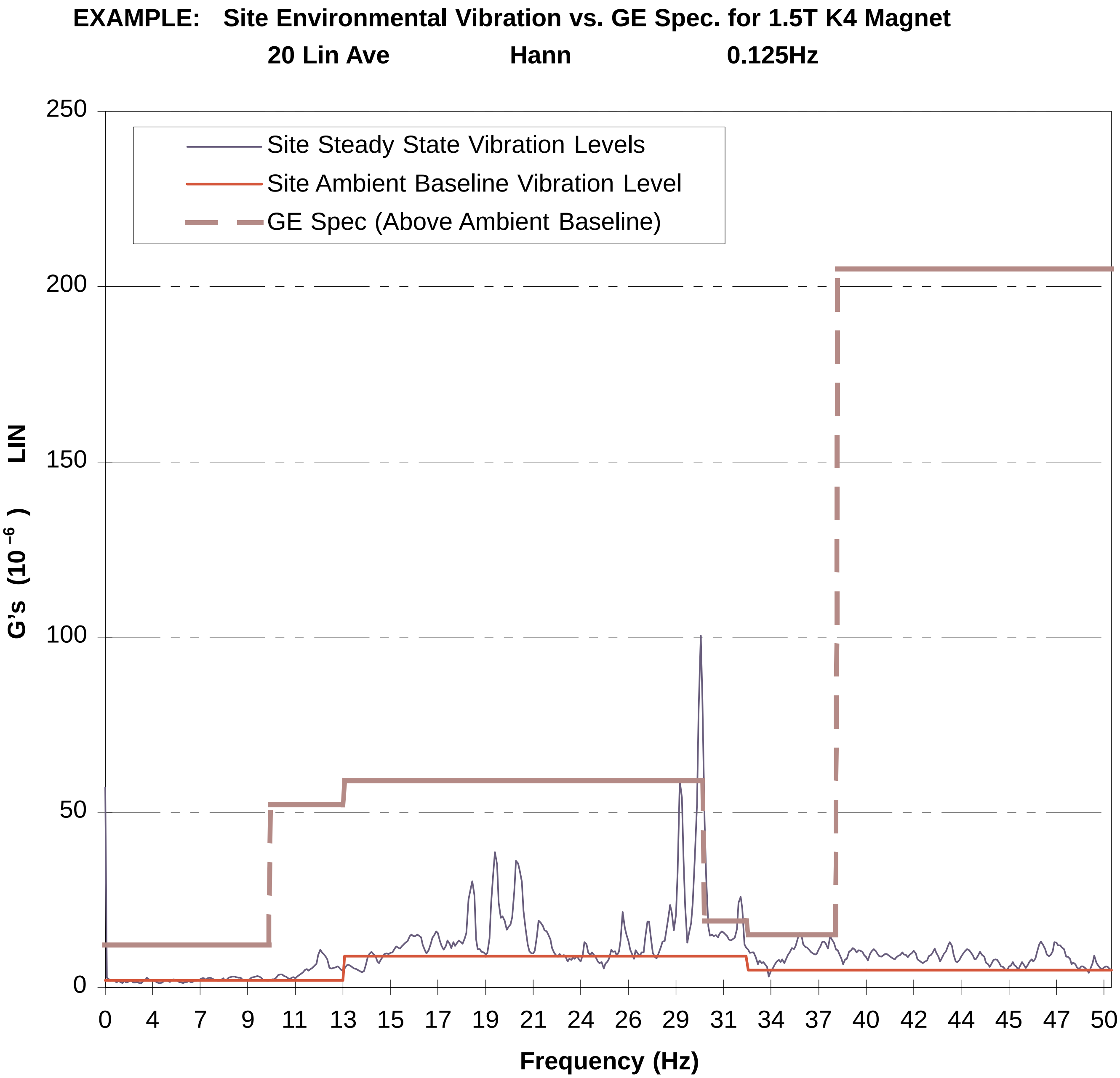
Any questions regarding test equipment requirements, test parameters, or general questions should be discussed with your GE Installation Specialist.

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



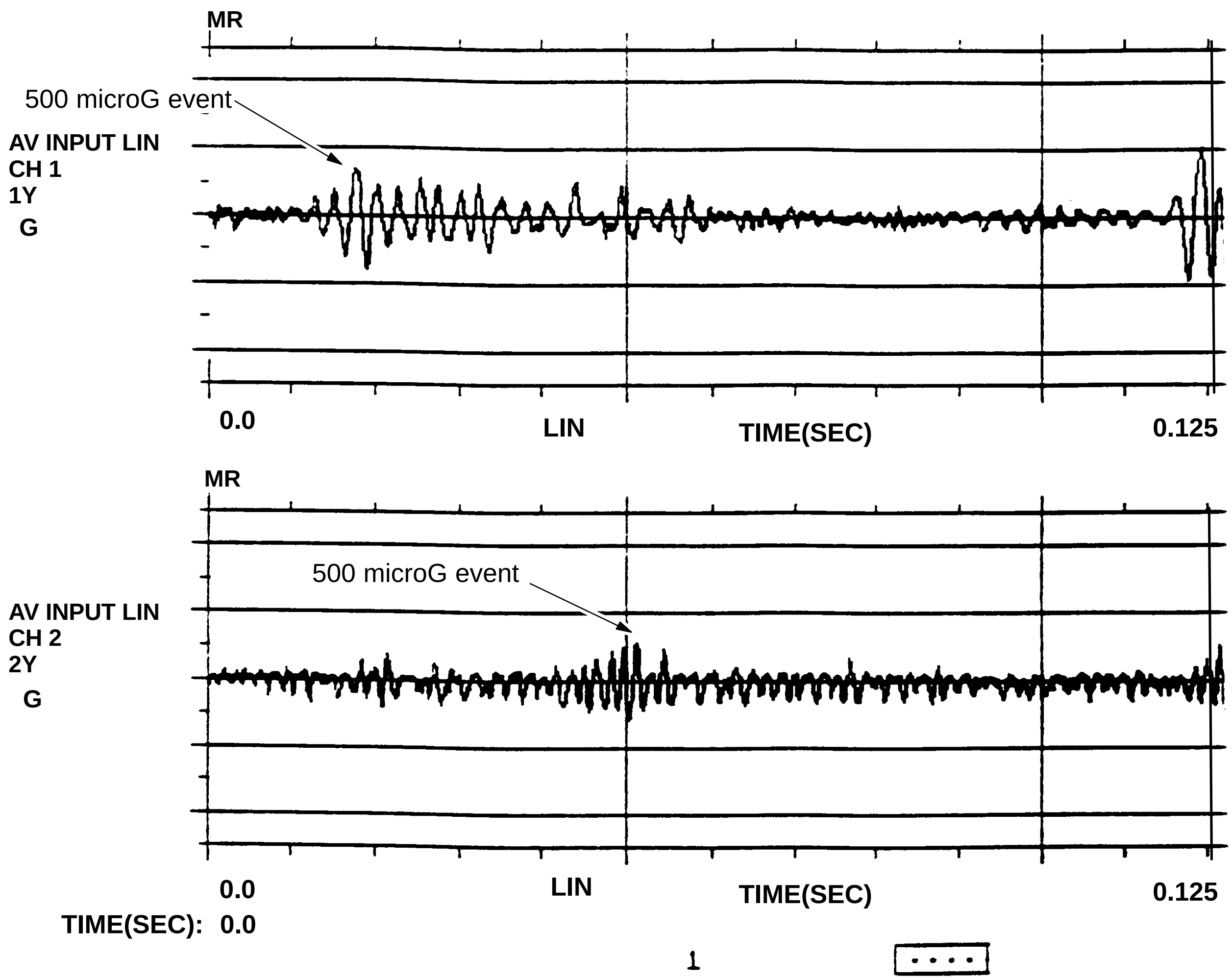
EXAMPLE SITE ENVIRONMENTAL VIBRATION
ILLUSTRATION A–1

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



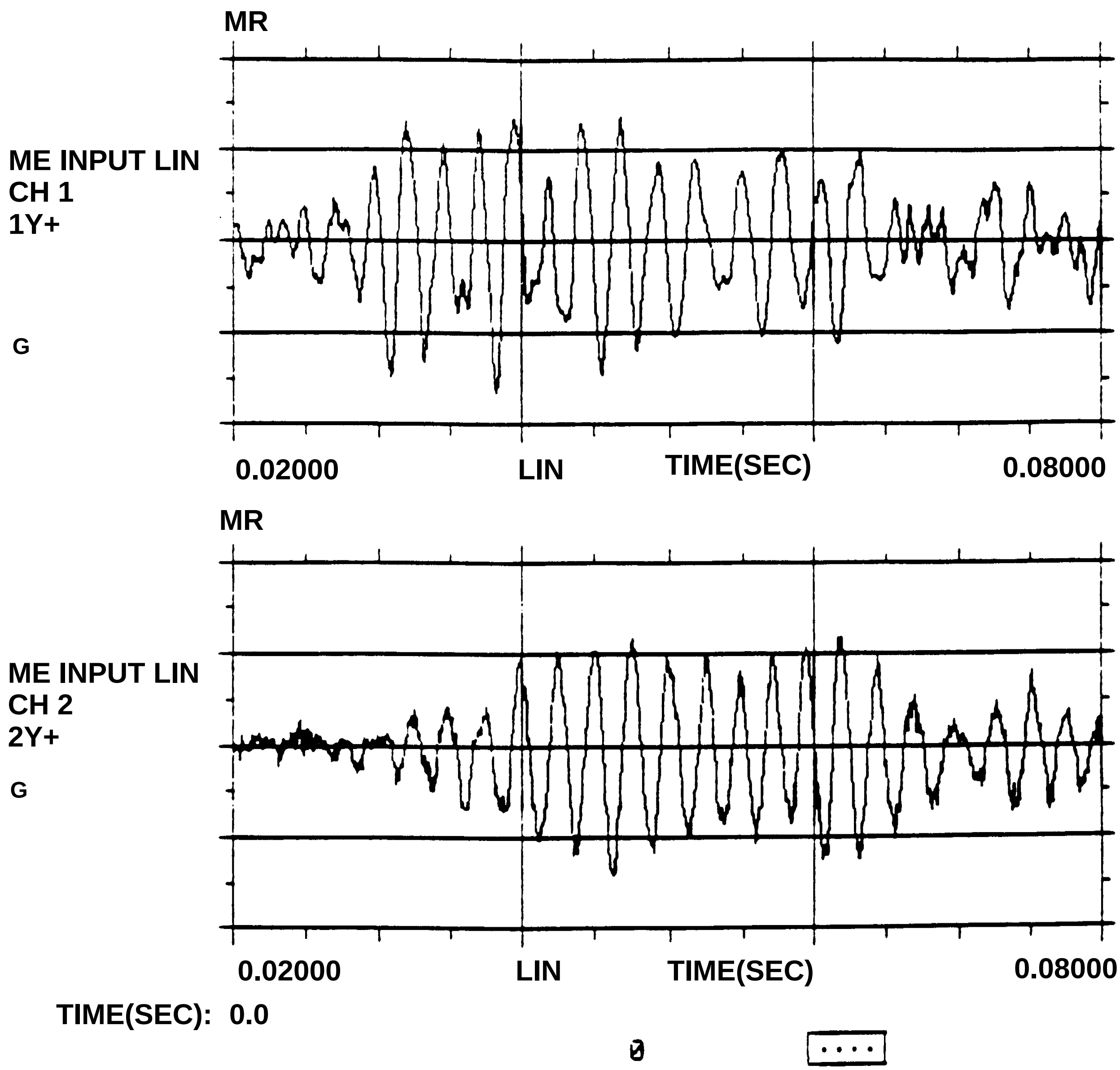
EXAMPLE SITE ENVIRONMENTAL VIBRATION VS. GE SPECIFICATION FOR 1.5T LCC MAGNET WITHOUT GRADIENT ISOLATION
ILLUSTRATION A–2

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY
ILLUSTRATION A–3

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY (ZOOM IN ON 500 microG TRANSIENT EVENT)
ILLUSTRATION A–4